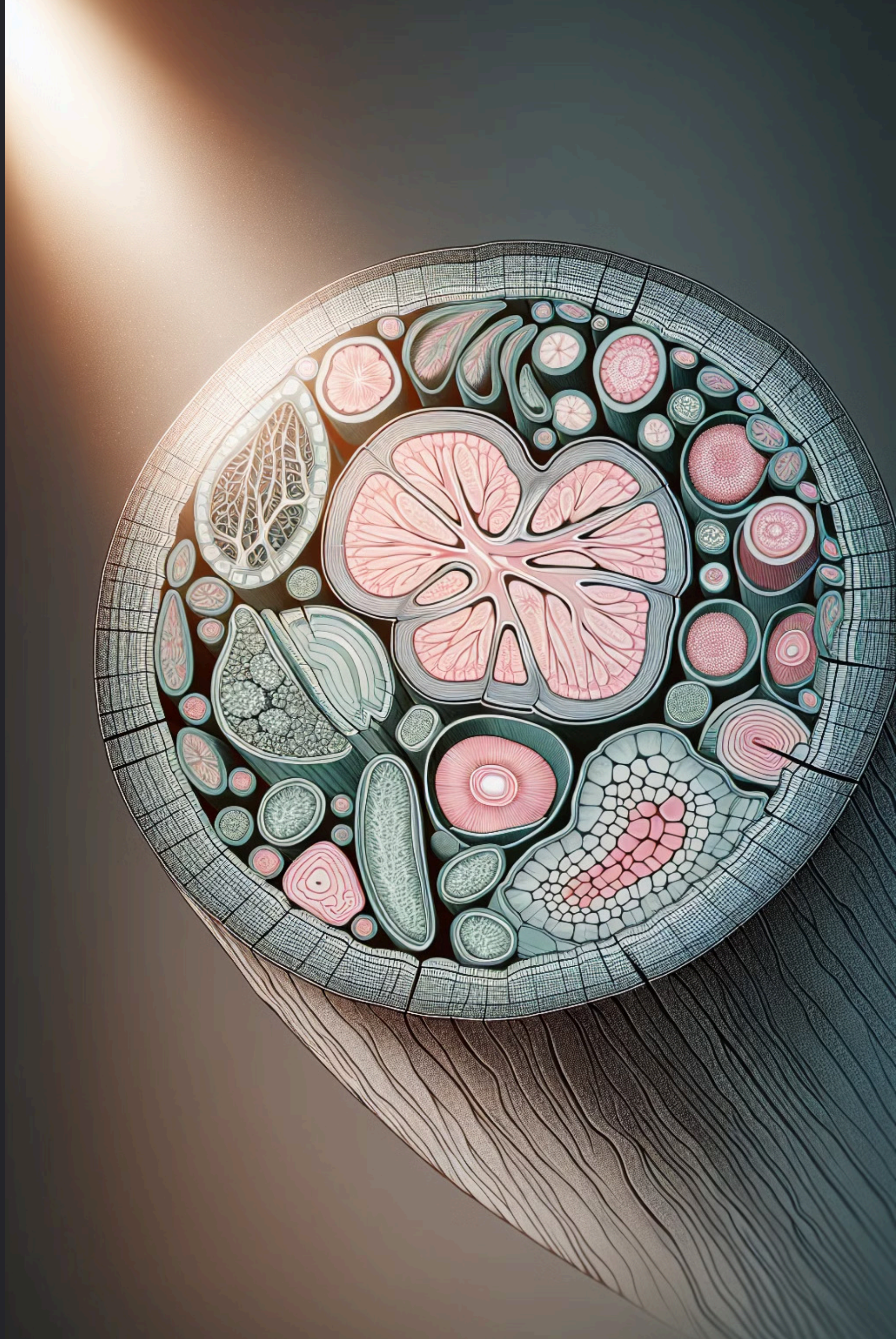
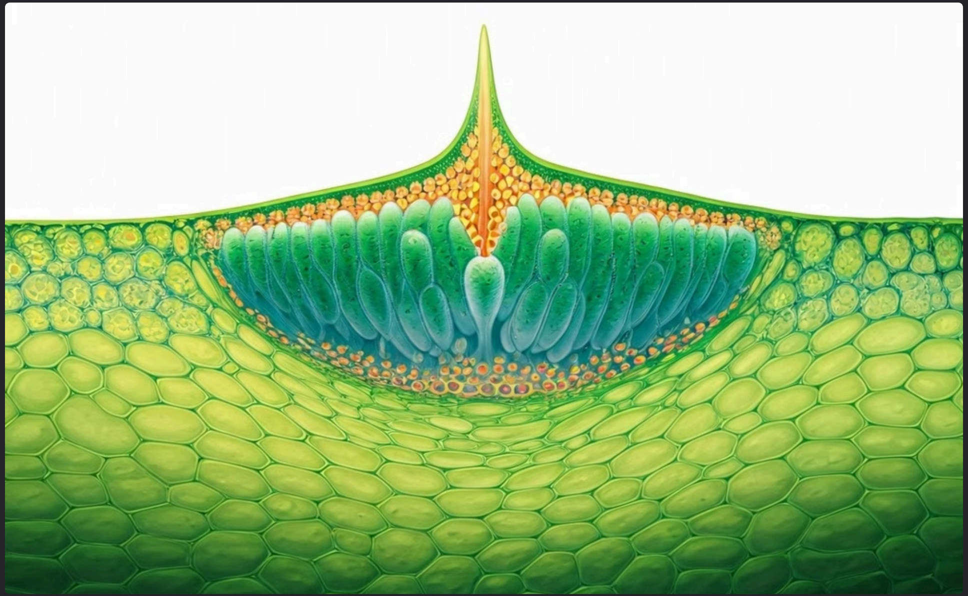
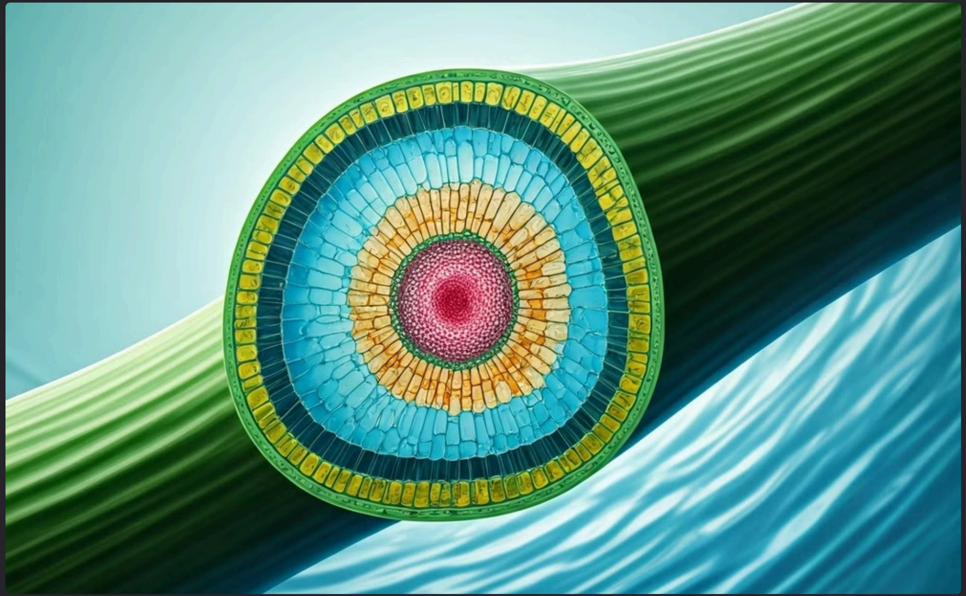
Tissues: Plant Tissues

This presentation will provide an overview of tissue types, structure, and function. Tissues are groups of similar cells that work together to perform a specific function. We will explore the different types of tissues found in plants and animals, as well as their unique characteristics and roles in the organism.



Introduction to Tissues





Definition of Tissues

Tissues are groups of similar cells that work together to perform a specific function. These cells are organized and coordinated, forming the building blocks of complex organisms.

Importance of Tissues

Tissues are essential for the structure and function of both plants and animals. They provide support, protection, and allow for specialized functions, contributing to the overall well-being of the organism.

Relationship Between Tissue, Organ, Organ System, and Organism

1. **Tissue**

   A group of similar cells working together to perform a specific function.

2. **Organ**

   A structure made up of different tissues that work together to perform a specific function.

3. **Organ System**

   A group of organs that work together to perform a specific function.

4. **Organism**

   A complete living being made up of different organ systems.

Plant Tissues Overview

Meristematic Tissue

Meristematic tissue is responsible for plant growth. It is composed of undifferentiated cells that can divide and differentiate into other types of cells. This tissue is found in specific locations within the plant, such as the tips of roots and stems.

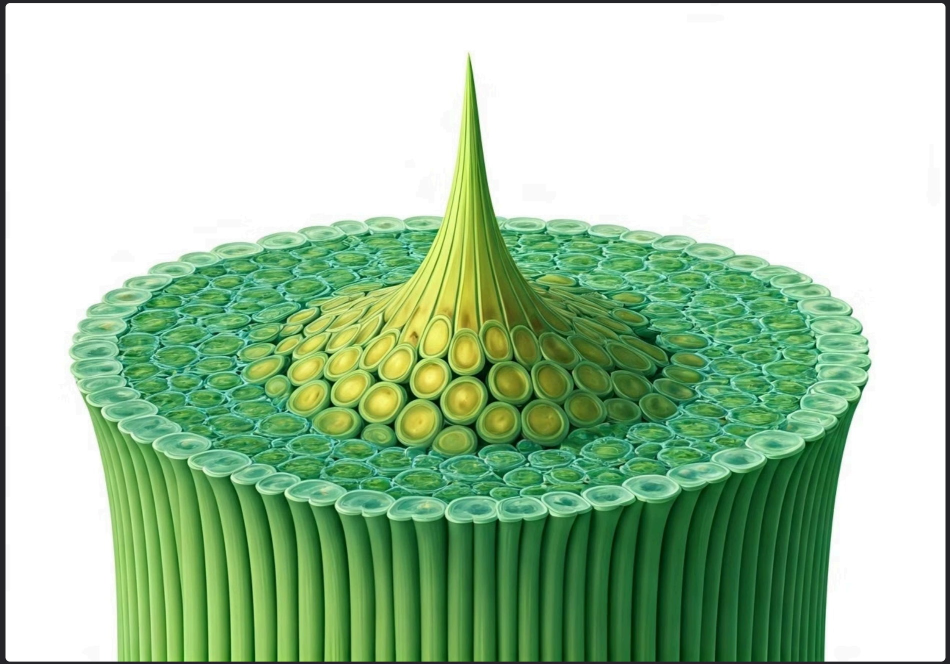
Permanent Tissue

Permanent tissue is composed of cells that have stopped dividing and have differentiated into specialized cells. These cells perform specific functions within the plant, such as photosynthesis, support, and transport.

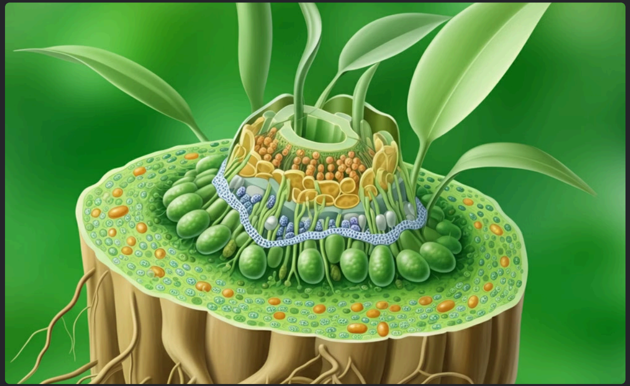
Meristematic Tissue (Meristem)

Meristematic tissue is a type of plant tissue that is responsible for growth. It is characterized by its actively dividing cells, which are responsible for increasing the length and width of the plant.

Meristematic tissue is found in specific locations within the plant, including the tips of roots, stems, and branches. These locations are known as apical meristems, and they are responsible for primary growth, which is the increase in length of the plant.

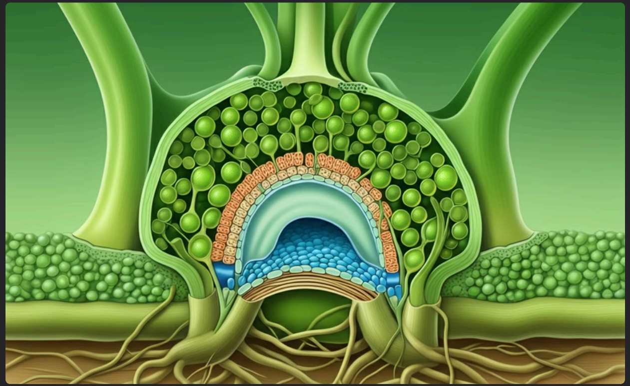


Characteristics of Meristematic Tissue



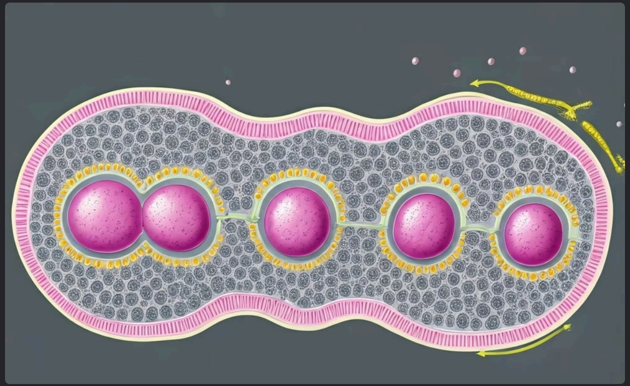
1. Small, Cubical Cells

Meristematic cells are small and cubical in shape. They have thin cell walls and large nuclei. These features allow for rapid cell division and growth.



2. Tightly Packed Cells

Meristematic cells have almost no vacuoles. This means that the cells are tightly packed together, leaving little space between them. This arrangement facilitates efficient communication and nutrient exchange.



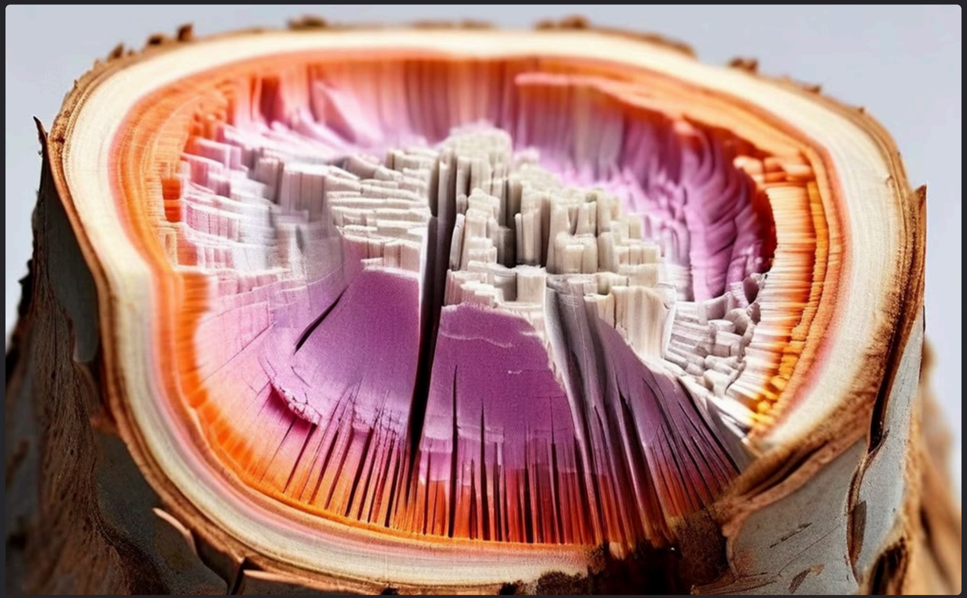
3. Continuous Cell Division

Meristematic tissue is characterized by continuous cell division. This process allows for the growth and development of the plant. The cells divide and differentiate into various permanent tissues, contributing to the plant's structure and function.

Types of Meristematic Tissue

Apical Meristem

Apical meristem is located at the tips of roots and stems. It is responsible for increasing the length of the plant. Apical meristem cells divide rapidly, producing new cells that differentiate into various tissues.



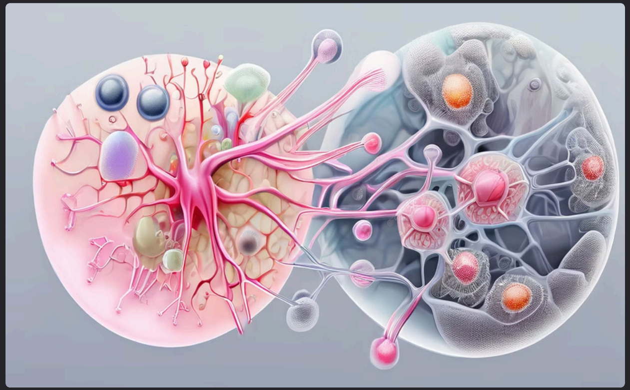
Lateral Meristem (Cambium)

Lateral meristem, also known as cambium, is located below the bark of woody plants. It is responsible for increasing the stem diameter. Cambium cells divide to produce new xylem and phloem cells, which transport water and nutrients throughout the plant.

Permanent Tissues Overview

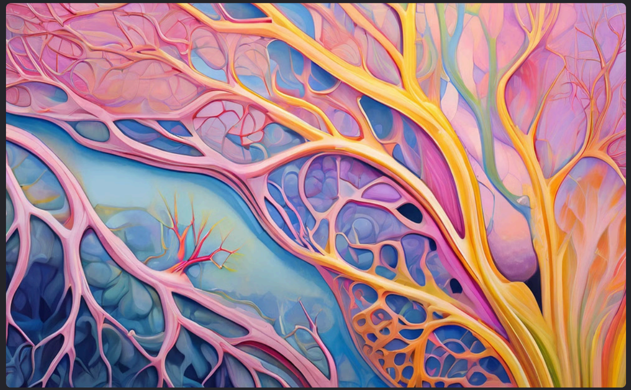
Protective Tissues

Protective tissues, such as the epidermis, protect the plant from external damage and regulate water loss.



Supporting Tissues

Supporting tissues, such as collenchyma and sclerenchyma, provide structural support and strength to the plant.



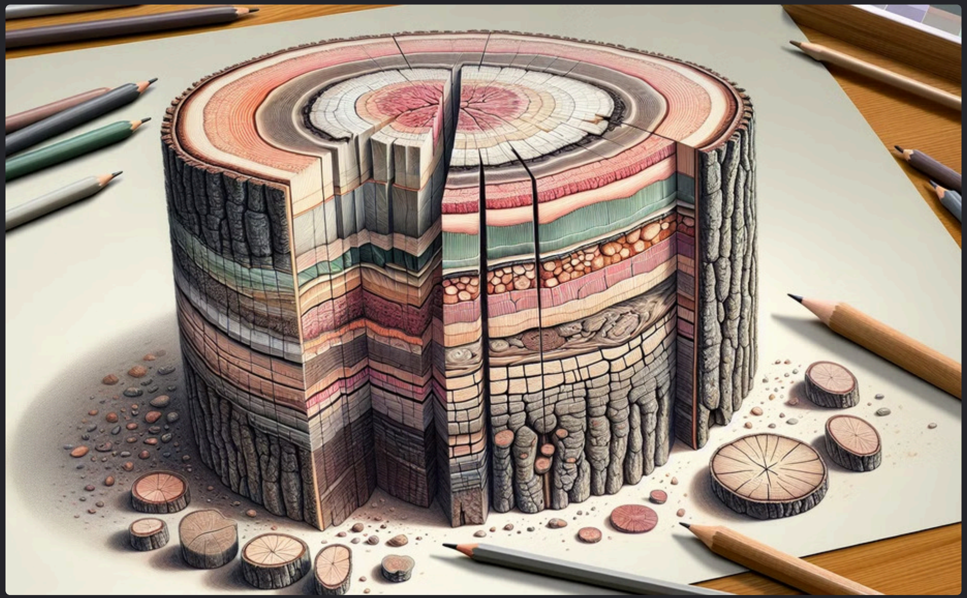
Conducting Tissues

Conducting tissues, such as xylem and phloem, transport water, nutrients, and sugars throughout the plant.

Protective Tissues

Epidermis

The epidermis is the outermost layer of cells that covers the plant's surface. It protects the plant from water loss and injury. The epidermis is covered by a waxy cuticle that helps to prevent water loss.



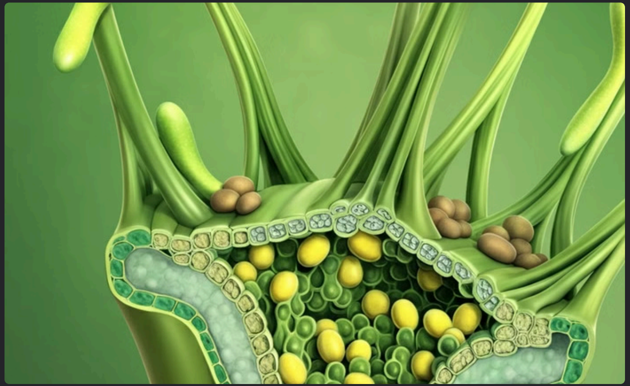
Cork

Cork is a protective tissue that forms in older stems and roots. It replaces the epidermis as the plant grows. Cork is made up of dead cells that are filled with air, which makes it a good insulator and helps to protect the plant from injury.

Supporting Tissues Overview

Parenchyma

Parenchyma is a type of simple permanent tissue found in all parts of the plant. Parenchyma cells have thin walls and a large central vacuole. They are responsible for storage, photosynthesis, and secretion.



Collenchyma

Collenchyma is another type of simple permanent tissue. It is found in the cortex of stems and leaves. Collenchyma cells are elongated and have thickened cell walls. They provide support and flexibility to the plant.

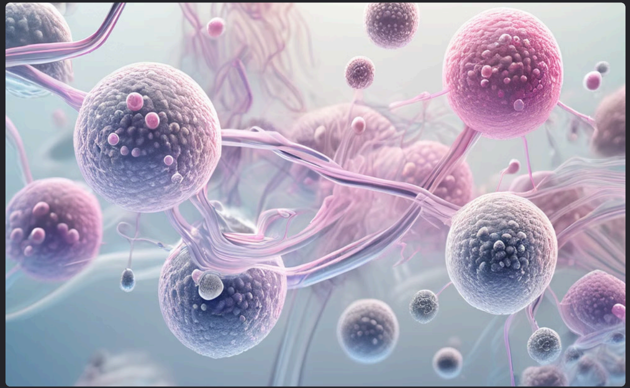
Sclerenchyma

Sclerenchyma is the third type of simple permanent tissue found in the vascular bundles of stems and roots. Sclerenchyma cells have thick walls and a rigid structure. They provide strength and support to the plant.

Parenchyma

Characteristics

Parenchyma cells are large and thin-walled. They often contain a single, large vacuole. This structure allows for efficient storage of water and nutrients.



Structure

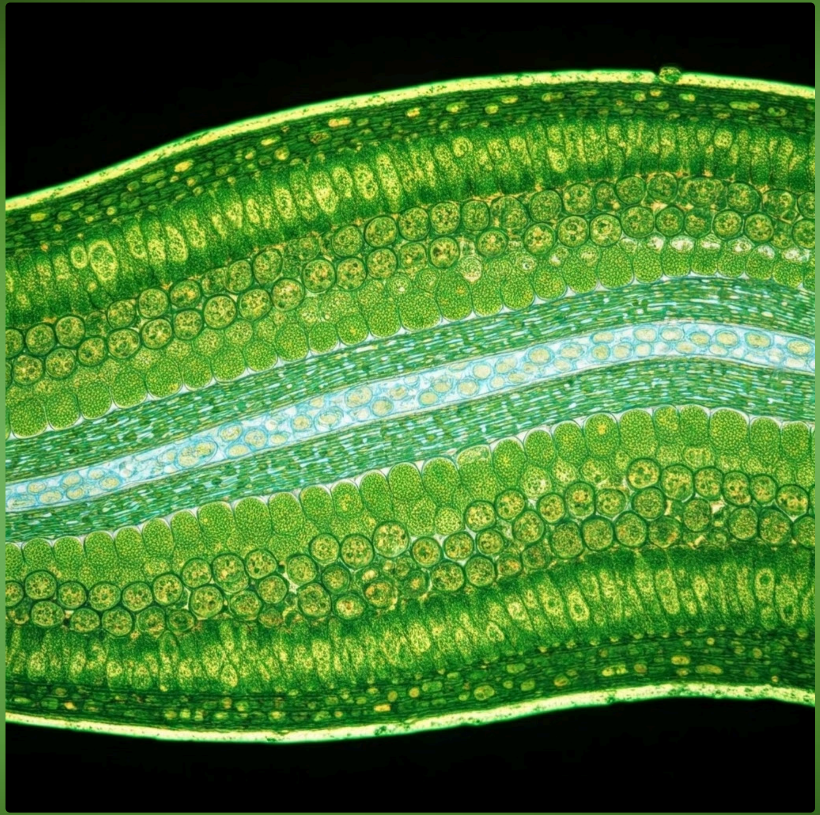
Parenchyma cells are often tightly packed together in a tissue, forming a continuous network that supports and provides structure to the plant.

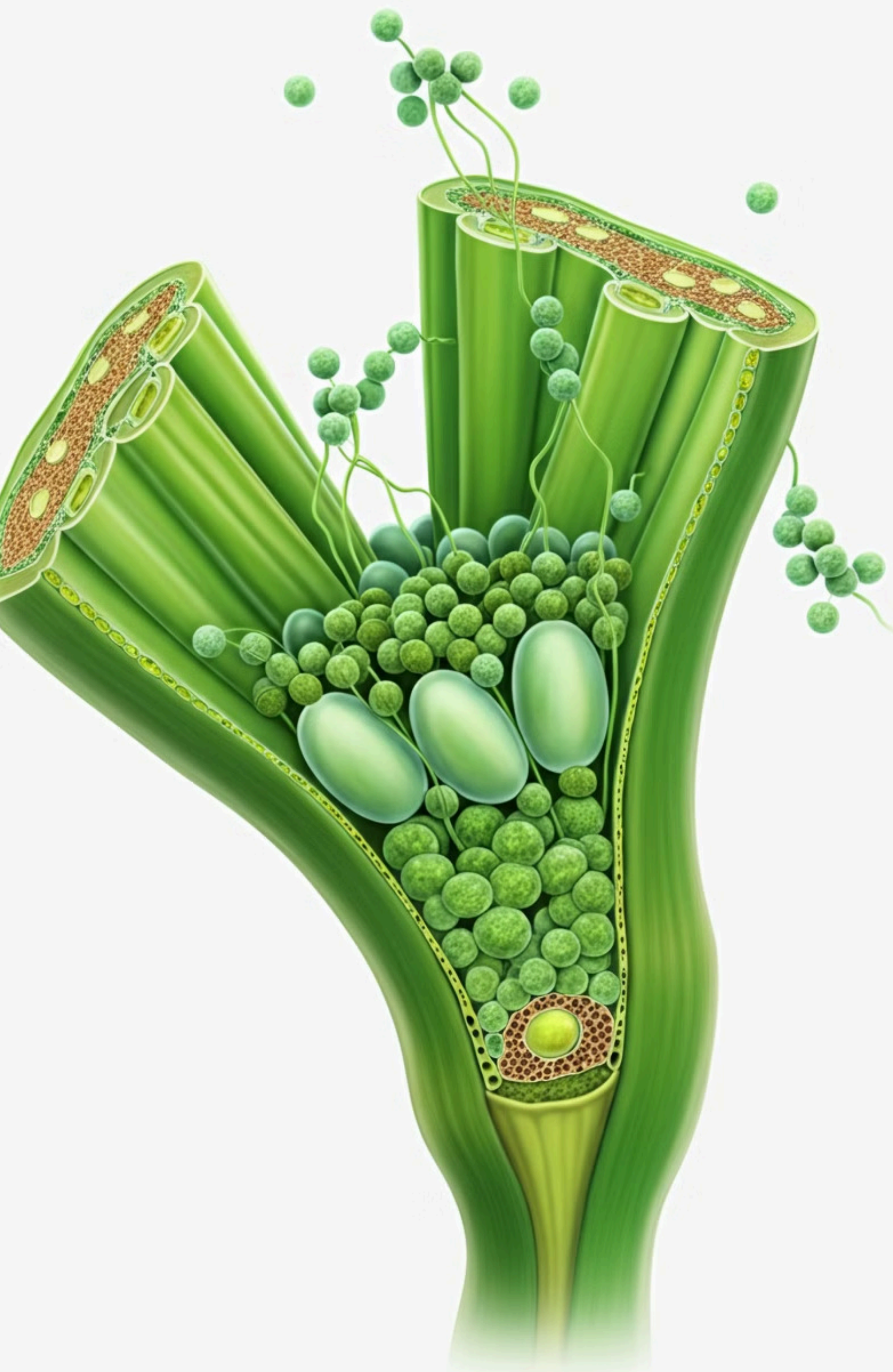
Function

Parenchyma cells play a crucial role in plant function. They store food, perform photosynthesis, and provide temporary support for the plant. These cells are essential for the plant's growth and survival.

Parenchyma (Visual)

Parenchyma cells are found in all parts of the plant, including the roots, stem, and leaves. They are responsible for a variety of functions, including photosynthesis, storage, and support. The image shows a cross-section of parenchyma cells in different parts of the plant.





Collenchyma

Characteristics

Collenchyma cells are elongated with thickened cell walls at the corners. These cells are living and have a flexible structure, allowing them to stretch and bend with the plant's growth.

Location

Collenchyma tissue is typically found in leaf stalks and below the epidermis of stems. This strategic placement allows it to provide support to the plant while still allowing for flexibility.

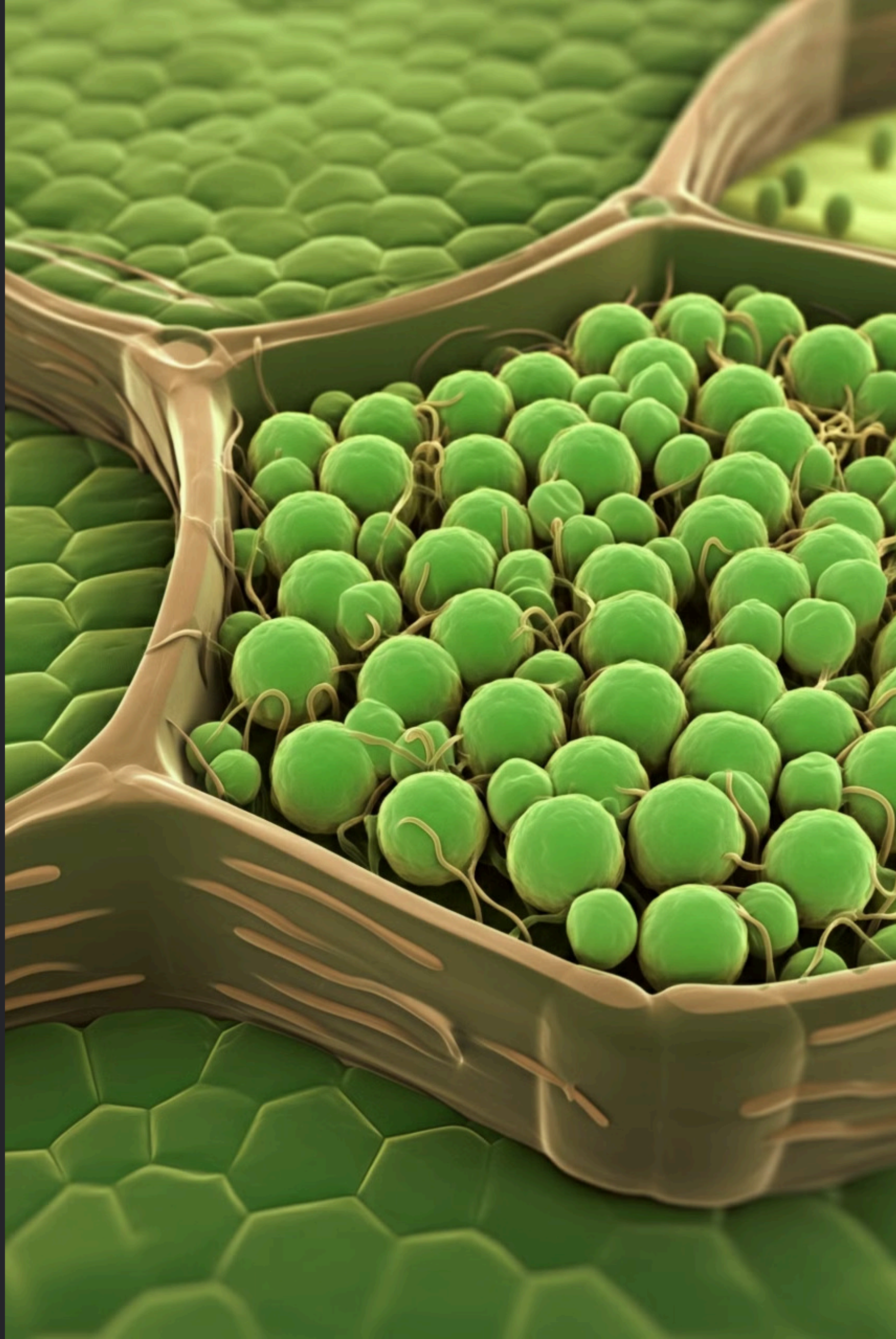
Function

Collenchyma's primary function is to provide support to the plant, particularly in young, growing stems and leaves. The thickened cell walls give the tissue its strength and flexibility, allowing it to withstand bending and stretching forces.

Collenchyma (Visual)

This diagram shows longitudinal and cross-sections of collenchyma tissue. Collenchyma cells are elongated and have thickened cell walls. The thickened cell walls are made of cellulose and pectin, which gives the cells their strength and flexibility.

Collenchyma tissue is found in the stems and leaves of plants. It provides support for the plant and helps to prevent the plant from wilting. Collenchyma tissue is also important for the growth and development of the plant.



Sclerenchyma

Characteristics

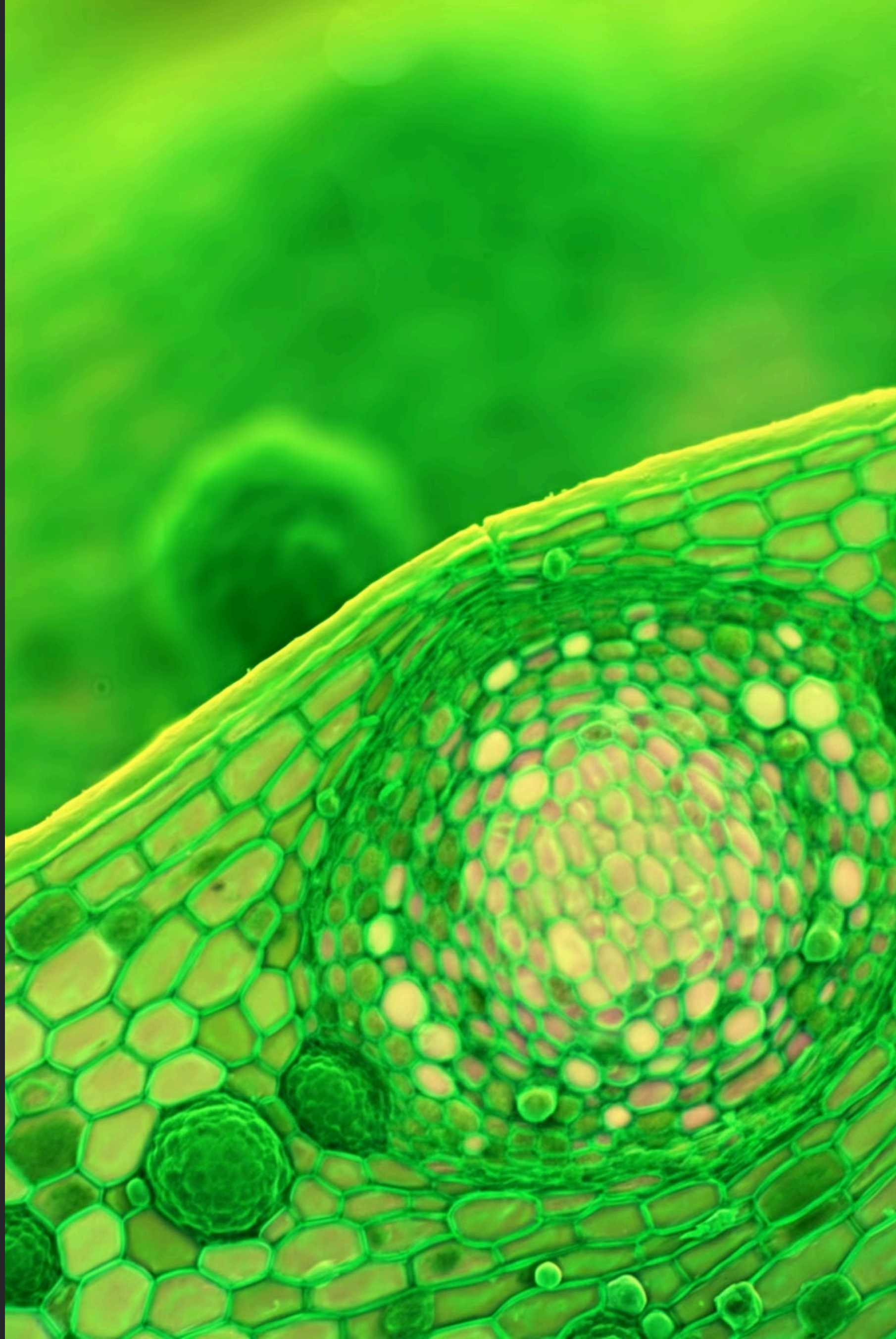
Sclerenchyma cells are long, narrow, and dead at maturity. They have thick, lignified walls, which provide strength and support to the plant. These cells are found in various parts of the plant, including stems, roots, and leaves.

Function

The primary function of sclerenchyma is to provide structural support and rigidity to the plant. The thick, lignified walls of these cells resist bending and compression, helping to maintain the plant's shape and prevent it from collapsing.

Location

Sclerenchyma cells are commonly found in the stems and veins of leaves. They also occur in the seed coats and fruit walls, providing protection and support to these structures.

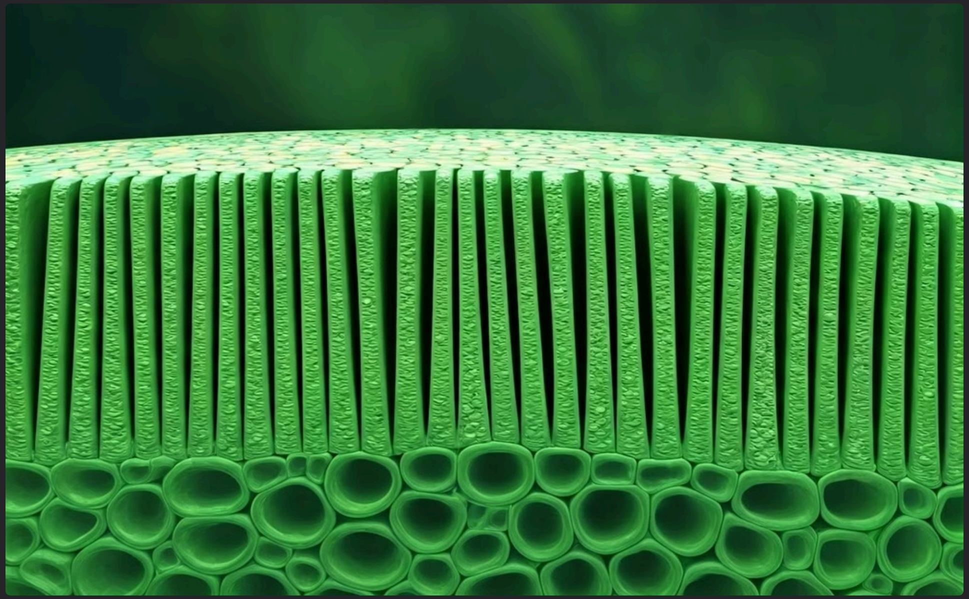


Sclerenchyma (Visual)

Sclerenchyma cells are dead at maturity and have thick, lignified cell walls. They provide structural support and rigidity to the plant. Sclerenchyma cells are found in various parts of the plant, including the stems, roots, and leaves. They are responsible for the strength and durability of these plant parts.

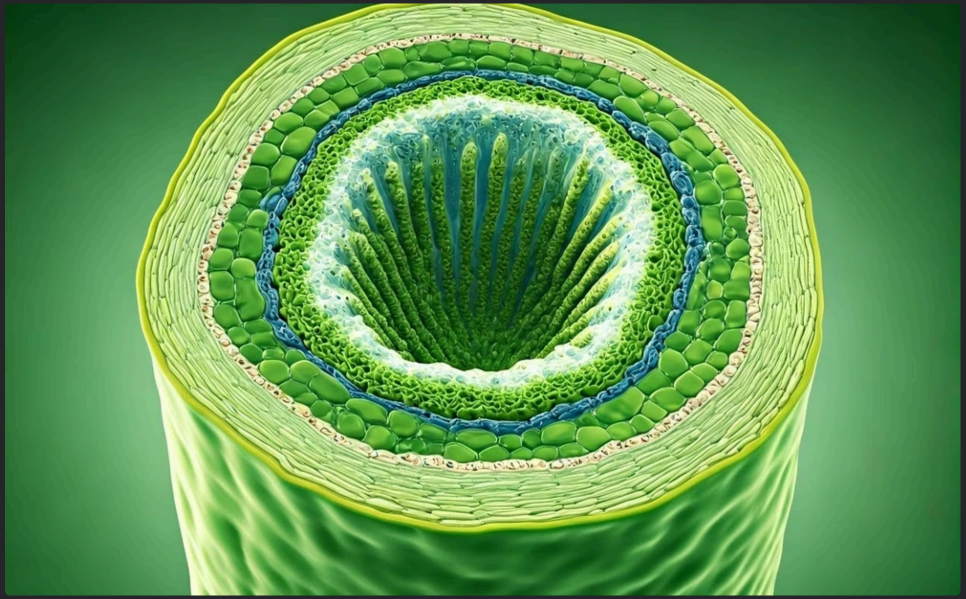
Sclerenchyma cells are often found in groups, forming a continuous network throughout the plant. This network helps to support the plant and prevent it from collapsing under its own weight. Sclerenchyma cells are also important for protecting the plant from herbivores and other environmental stresses.

Conducting Tissues Overview



Xylem

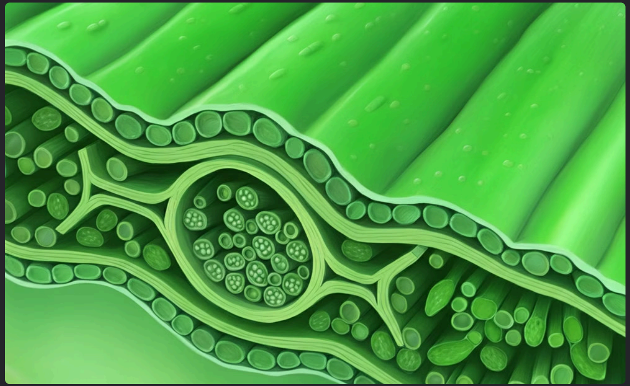
Xylem is one of the two types of conducting tissues found in plants. It is responsible for transporting water and dissolved minerals from the roots to the rest of the plant. Xylem is made up of dead cells that are joined end to end, forming long tubes. These tubes are strengthened by lignin, a rigid substance that gives the xylem its structural support.

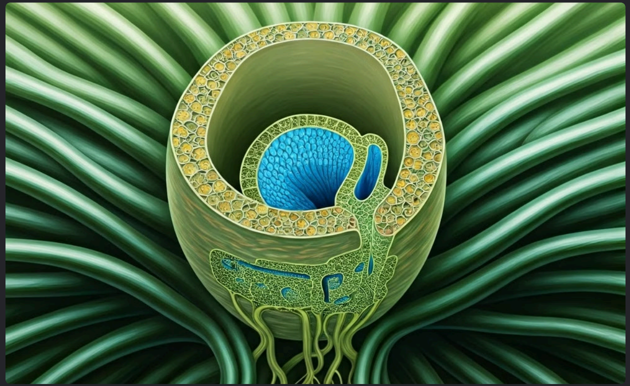


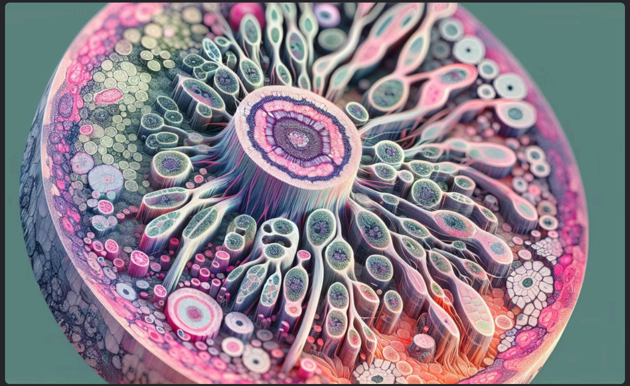
Phloem

Phloem is the other type of conducting tissue in plants. It is responsible for transporting sugars and other organic compounds produced during photosynthesis from the leaves to the rest of the plant. Phloem is made up of living cells that are connected by sieve plates, which are porous structures that allow for the passage of fluids. Phloem also contains companion cells, which provide support and regulate the activity of the sieve tubes.

Xylem







Elongated Cells

Xylem cells are elongated and have thick walls, forming tubular passages. These cells are dead at maturity, but their cell walls remain, providing structural support and allowing for efficient water transport.

Water Transport

Xylem is responsible for transporting water and dissolved minerals from the roots to other parts of the plant. This process is essential for plant growth, photosynthesis, and overall survival.

Components

Xylem is composed of several types of cells, including tracheids, vessels, xylem parenchyma, and xylem fibers. Each component plays a specific role in the structure and function of xylem tissue.

Tracheids

Tracheids Description

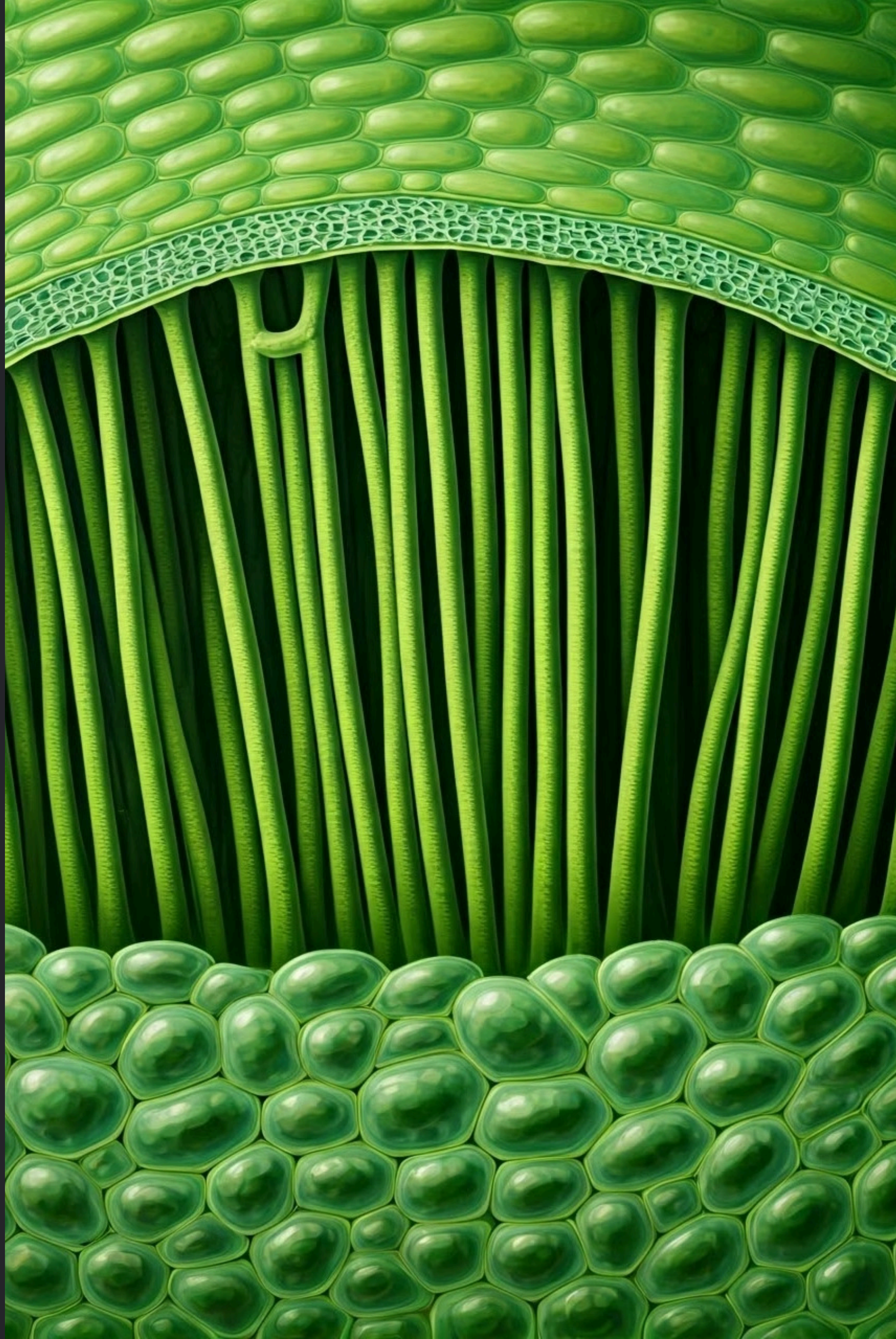
Tracheids are elongated, dead cells with thick walls and tapered ends. They form long, continuous tubes that transport water and dissolved minerals from the roots to the rest of the plant.

Tapered Ends and Pits

The tapered ends of tracheids overlap, forming pits that allow water to flow from one cell to the next. These pits are also important for providing structural support to the xylem.

Importance in Plants

Tracheids are found in all vascular plants, but they are particularly important in gymnosperms, which lack vessels.



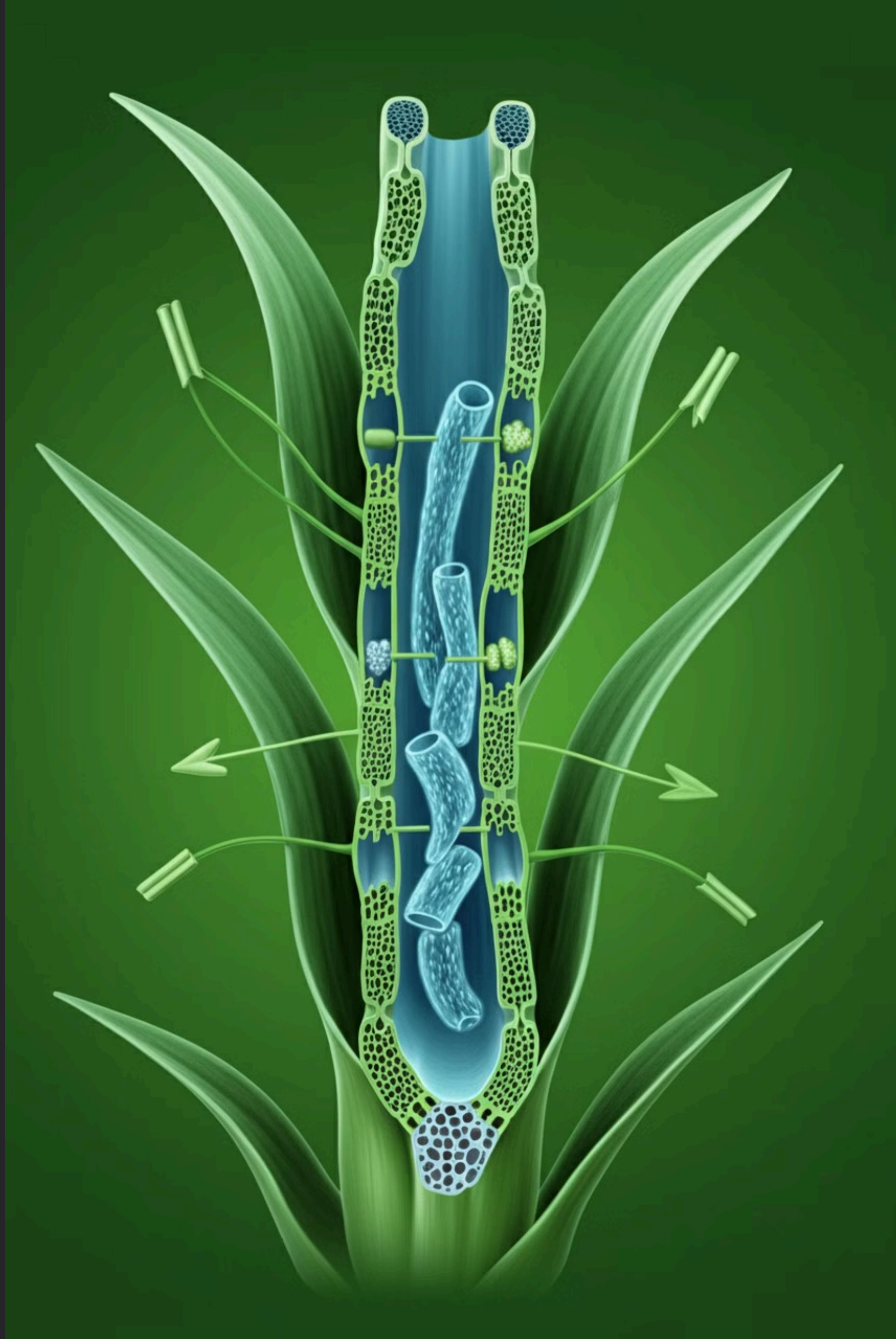
Xylem Vessels

Xylem Vessels

Xylem vessels are another component of the xylem, along with tracheids. Vessels are also elongated, dead cells, but unlike tracheids, they are wider and have perforated end walls. These perforations allow for the uninterrupted flow of water through the vessels, making them more efficient at transporting water.

Angiosperms vs. Gymnosperms

Vessels are found in angiosperms, flowering plants. They are typically wider than tracheids, contributing to the increased efficiency of water transport in angiosperms. This improved efficiency is crucial for the growth and survival of angiosperms, enabling them to thrive in diverse environments.



Xylem Parenchyma

Support and Storage Functions

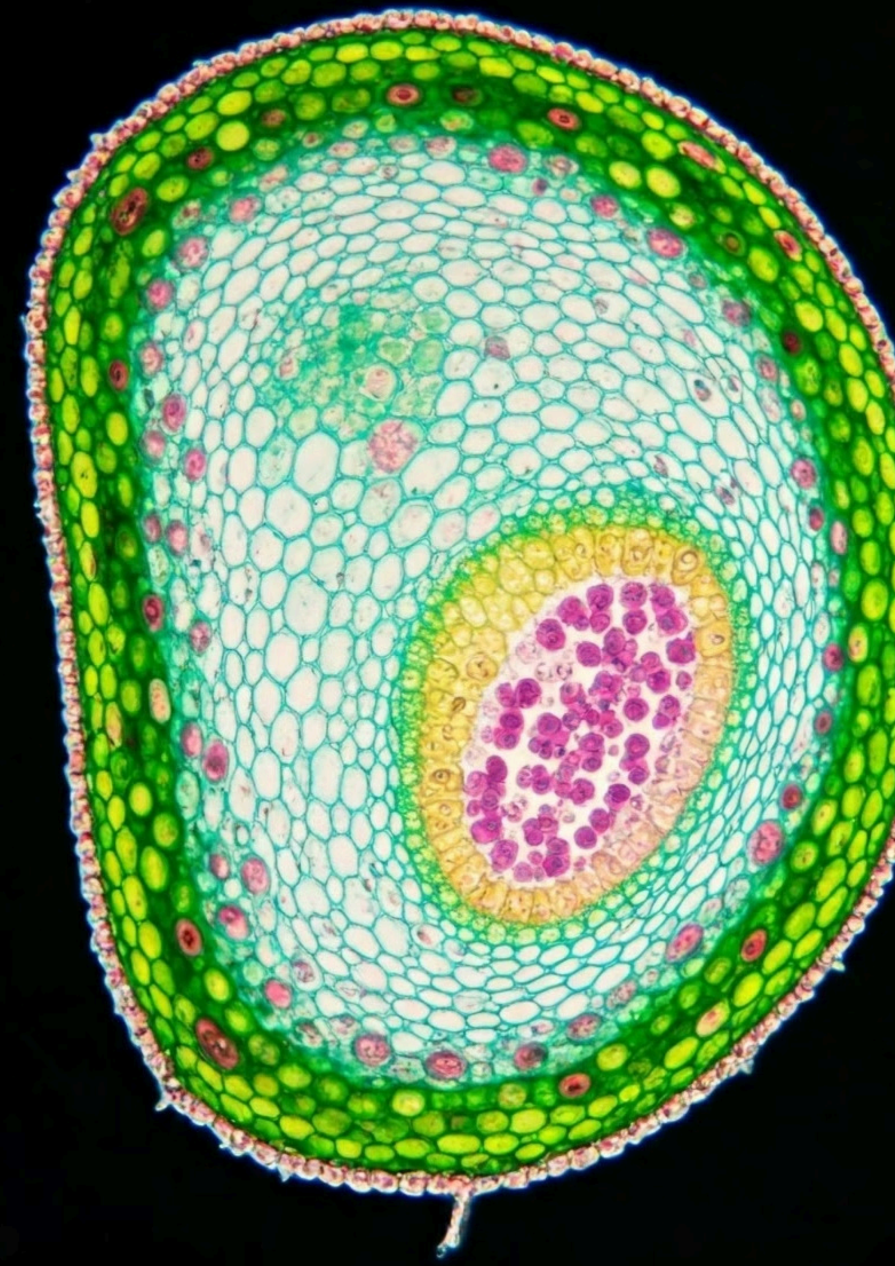
Xylem parenchyma cells provide support and storage functions in xylem tissue. They are essential for maintaining the structural integrity of the plant and storing essential nutrients.

Role in Nutrient Exchange

These cells are connected by plasmodesmata, enabling communication and nutrient exchange between cells. This interconnected network is vital for the overall health and function of the plant.

Regulation and Signaling

Xylem parenchyma cells play a key role in regulating water potential, storing water, and producing important signaling molecules for plant growth and development.

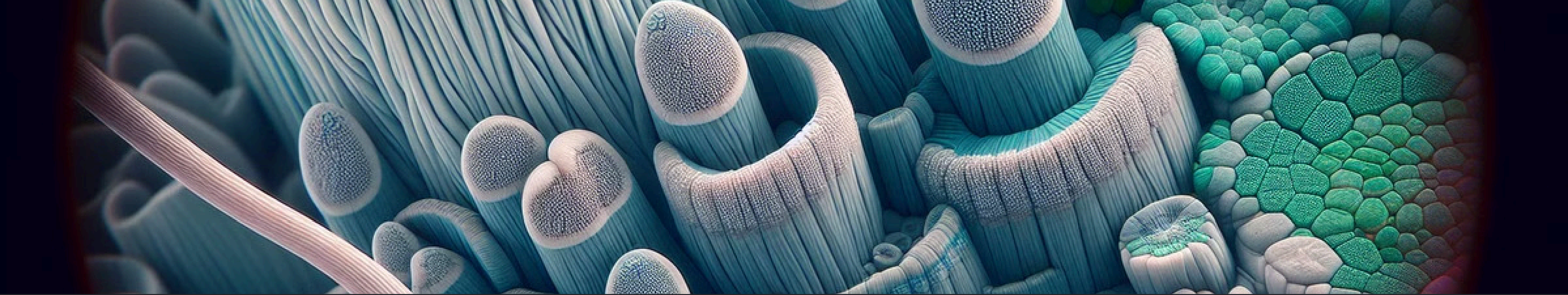


Xylem Fibre

Xylem fibres are another type of cell found in xylem tissue. They are elongated and have thick, lignified cell walls.

These cells are dead at maturity, and their primary function is to provide support and strength to the xylem tissue.

Xylem fibres are important for maintaining the structural integrity of the plant, especially in tall plants that need to withstand strong winds or gravity.



Phloem

Characteristics

Phloem cells form a passage for the movement of food. They are responsible for transporting sugars and other nutrients from the leaves to storage organs and other parts of the plant.

Function

Phloem is essential for the survival of plants. It allows them to distribute nutrients throughout their bodies, ensuring that all parts of the plant have access to the energy they need to grow and thrive.

Components

Phloem is composed of several different types of cells, including sieve tubes, companion cells, phloem parenchyma, and phloem fibers. Each of these cell types plays a specific role in the transport of food.

Phloem (Visual)

Phloem is a type of vascular tissue in plants that is responsible for transporting sugars and other organic molecules from the leaves to other parts of the plant. The phloem is made up of several types of cells, including sieve tubes and companion cells.

Sieve tubes are long, thin cells that are connected end-to-end to form a continuous tube. The sieve tubes have pores in their cell walls, which allow for the passage of sugars and other molecules. Companion cells are smaller cells that are located next to the sieve tubes. They provide the sieve tubes with energy and other essential nutrients.

Transportation of Sugars

Phloem is responsible for transporting sugars and other organic molecules from the leaves to other parts of the plant.

Sieve Tubes

Sieve tubes are long, thin cells connected end-to-end to form a continuous tube. They have pores in their cell walls for the passage of sugars and molecules.

Companion Cells

Companion cells provide energy and essential nutrients to the sieve tubes, supporting their function in transporting substances throughout the plant.

Vascular Bundles

1 **Xylem and Phloem**

Xylem and phloem are the two main types of vascular tissue in plants. They work together to form vascular bundles, which are responsible for transporting water, nutrients, and food throughout the plant.

2 **Transport**

Xylem transports water and dissolved minerals from the roots to the rest of the plant. Phloem transports sugars produced by photosynthesis from the leaves to other parts of the plant.

3 **Veins**

Vascular bundles are visible as veins in leaves. They provide a network of pathways for the transport of essential substances, allowing the plant to grow and thrive.

Summary of Plant Tissues

Meristematic Tissue

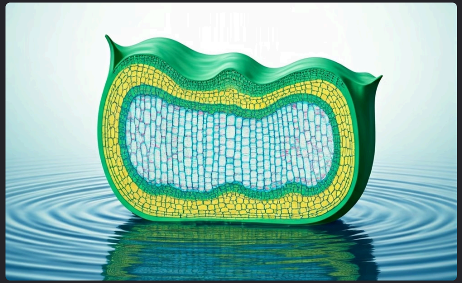
Meristematic tissue is responsible for plant growth. It's made up of undifferentiated cells that can divide and differentiate into other types of cells.

Protective Tissue

Protective tissue covers the outer surface of plants and protects them from damage and infection. It's made up of cells that are tightly packed together and often have a waxy coating.

Supporting Tissue

Supporting tissue provides structural support for plants. It's made up of cells that are thick-walled and often have a rigid structure.



Conducting Tissue

Conducting tissue transports water, nutrients, and sugars throughout the plant. It's made up of specialized cells that form tubes or vessels.